

AE 17: Multicollinearity

Rail Trail

Driver: _____, Reporter: _____, Gopher: _____

! Important

- Open [RStudio](#) and create a subfolder in your AE folder called “AE-17”.
- Go to the [Canvas](#) and locate your AE-17 assignment to get started.
- Upload the `ae-17.qmd` and `rail_train.csv` files into the folder you just created. The `.qmd` and PDF responses are due in Canvas. You can check the due date on the Canvas assignment.

Packages + data

```
library(tidyverse)
library(ggformula)
library(GGally)
library(broom)
library(knitr)
library(mosaic)
library(mosaicData)

rail_train <- read_csv("rail_train.csv")
```

The data for this AE is based on data from the Pioneer Valley Planning Commission (PVPC) and is included in the **mosaicData** package. The PVPC collected data for ninety days from April 5, 2005 to November 15, 2005. Data collectors set up a laser sensor, with breaks in the laser beam recording when a rail-trail user passed the data collection station. More information can be found [here](#).

Analysis goal

The goals of this activity are to:

- Detect and remedy multicollinearity in our data

Exercise 0

Fit a linear model to predict `volume` from ALL of the other predictors. The resulting model is called the “full model”.

Exercise 1

Create a new data frame called `rail_trail_new` with two new columns, `hightemp_c` and `avgtemp_c`. These two new columns should store the `hightemp` and `avgtemp` measured in Celsius rather than Fahrenheit. Note that the formula to convert between Fahrenheit and Celsius is $C = (F - 32) \times \frac{5}{9}$. Make sure `hightemp` and `avgtemp` are still in your data set.

Exercise 2

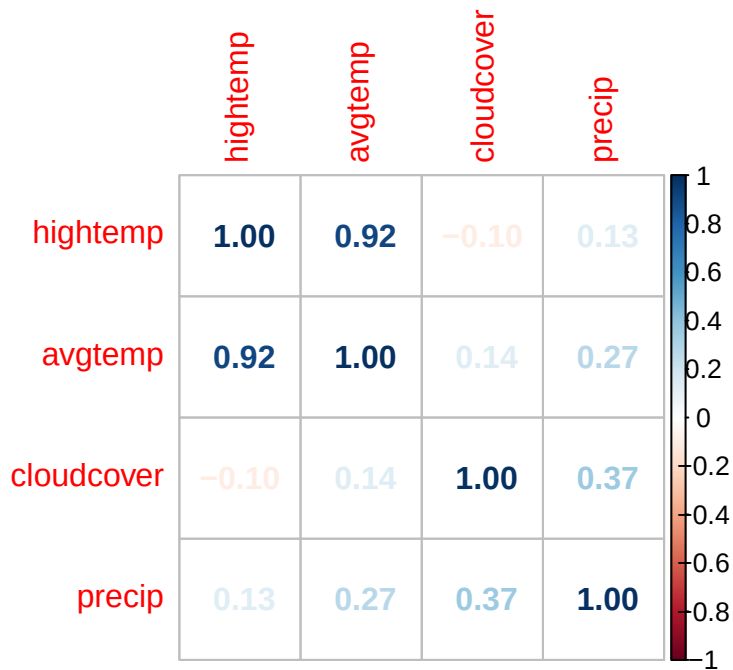
Try to refit the full model on your data. Do you notice anything odd about your coefficients? If so, what do you think is happening?

Exercise 3

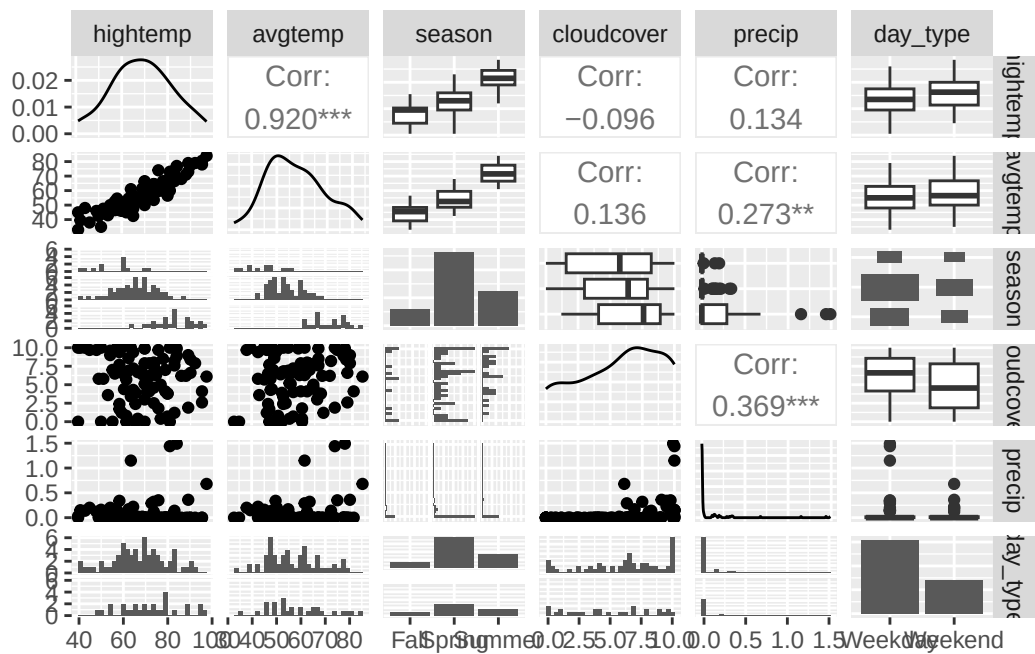
Add a comment next to each line of code explaining what that line of code is doing.

```
library(corrplot)

rail_trail |> # Take the rail_trail data set and then
  select(-volume) |> # What do I do
  select(where(is.numeric)) |> # What do I do
  cor() |> # What do I do
  corrplot(method="number") # What do I do
```



```
rail_train |>
  select(-volume) |> # What do I do
  ggpairs()
```



Exercise 4

Based on the output from the previous exercise, do you have reason to believe that our data is exhibiting multicollinearity?

Exercise 5

Pipe the model you fit in Exercise 0 (i.e. not including the variable measured in Celcius) into the function `vif`. Are any of the vif's concerning? If so, which ones.

Exercise 6

Fit two models:

- `m1`: use all predictors except for `avgtemp`
- `m2`: use all predictors except for `hightemp`
- `m3`: average `avgtemp` and `hightemp` together and use it instead of either `avgtemp` and `hightemp`

Exercise 7

Use `glance` to compute the R^2_{adj} , AIC, and BIC for both of these models. Which should we choose?

To submit the AE

! Important

- Render the document to produce the PDF with all of your work from today's class.
- Upload your QMD and PDF files to the Canvas assignment.